Lab # 3

File/Directory Permission and Process Management

I. File/Directory Permission

Type			
d (directory)	4 2 1 R w x (read) (write) (execute)	4 2 1 r w x (read) (write) (execute)	4 2 1 r w x (read) (write) (execute)
	4+2+1	4+0+1	000
	rwx	r-x	---
	owner	group	other

Linux (and almost all other Unixish systems) have three user classes as follows:

- User (u): The owner of file
- Group (g): Other user who are in group (to access files)
- Other (o): Everyone else

You can setup following mode on each files. In a Linux and UNIX set of permissions is called as mode:

- Read (r)
- Write (w)
- Execute (x)

You can use octal number to represent mode/permission:
- r: 4
- w: 2
- x: 1

For example, for file owner you can use octal mode as follows. Read, write and execute (full) permission on a file in octal is

0+r+w+x = 0+4+2+1 = 7

Only Read and write permission on a file in octal is

0+r+w+x = 0+4+2+0 = 6

Only read and execute permission on a file in octal is

0+r+w+x = 0+4+0+1 = 5

Permissions	**Symbolic**	**Binary**	**Octal**
read, write, and execute	rwx	111	7
read and write	rw-	110	6
read and execute	r-x	101	5
Read	r--	100	4
write and execute	-wx	011	3
Write	-w-	010	2
Execute	--x	001	1
no permissions	---	000	0

A) Working with Users

i) <u>Adding user:</u>

Login as root.

$ useradd <username>

Example: $ useradd omer

$ tail -5 /etc/passwd

$ id omer

ii) <u>Adding user with specific full name:</u>

$ useradd –c <userfullname> <username>

Example: $ useradd –c ‘Hadi Khan’ Hadi

$ tail -5 /etc/passwd

$ id Hadi

iii) <u>Creating user with account expiration date</u>

$ useradd -e ‘YYY-MM-DD’ <username>

Example: $ useradd -e ‘2024-12-31’ faisal

$ tail -5 /etc/passwd

$ id faisal

iv) Adding user in a group:

$ useradd –g <groupname> <username>

Example: $ useradd –g Students Hadi

$ tail -5 /etc/passwd

$ id Hadi

iv) Changing password:

$ passwd <username>
 Example: $ passwd Hadi

v) Loacking/unlocking password:

$ passwd –l <username>
 Example: $ passwd –l Hadi

$ passwd –u <username>
 Example: $ passwd –u Hadi

vi) Deleting user:

$ userdel <username>
 Example: $ del –g Students faisal
 $ tail -5 /etc/passwd
 $ id faisal

B) Working with Group

i) Adding Group:

Login as root.
$ groupadd <groupname>
Example: $ groupadd students
 $ tail -5 /etc/group

ii) Adding Group with specific Group ID:

$ groupadd –g <groupID(Numeric)> <groupname>
Example: $ groupadd –g 1009 OSstudents
 $ tail -5 /etc/group

iii) Changing Group ID/Group Name:

$ groupmod –g <newgroupID(Numeric)> <groupname>
Example: $ groupadd –g 1008 OSstudents
 $ tail -5 /etc/group

$ groupmod –n <newgroupname)> <groupname>
Example: $ groupadd –n OSLabStudents OSstudents
 $ tail -5 /etc/group

iv) Adding users to Group

$ usermod –aG <groupname> <username>
Example: $ groupadd –aG OSLabStudents omer
 $ id omer

v) Removing users from Group

$ gpasswd --delete <groupname> <username>
Example: $ gpasswd –-delete OSLabStudents omer

$ id omer

vi) Removing Group

$ groupdel <groupname>

Example: $ groupdel –-delete OSLabStudents

$ tail -5 /etc/group

C) File/Directory permissions

i) Changing owner

$ chown <newowner>**:**<newgroup> <directoryname/filename>

Example: $mkdir testdir

$ chown bilal:students testdir

$ ls –l

ii) Changing rights

$ chmod <u+rwx, g+rwx, o+rwx> <directoryname/filename>

Example: chmod 770 testdir

$ ls -lh

For example, if a text file has 666 permissions, it grants read and write permission to everyone. Similarly a directory with 777 permissions, grants read, write, and execute permission to everyone.

D) Process Management:

How to see processes

How to kill processes

i) Reviewing process

$ ps

What processes are running

$ ps ax

Show all-extended processes running

$ ps aux

Show processes running and user information

$ pstree

Show processes in tree structure.

ii) Killing Process

$ kill <pid>

Example: $ kill 2598

Kill Process immediately
$ kill -9 <pid>
Example: $ kill -9 2598

Kill Process with name
$ pkill <processname>
Example: $ pkill gcalctool

To kill processes with name.
$ pkillall <processname>
Example: $ killall gcalctool

iii) System Monitoring Commands

To show system time when it is open , Open from how many hours, User login and Load average of processes
$ uptime

System monitoring command, Uptime info, Processes refresh after every 2 minute
$ top
System monitoring command

Processes update after every 2 second.
$ watch uptime

Lab Tasks

1.Create two groups webProject and SQA.

2.Create two users , user1 and user2 and add them in a group webProject and SQA respectively.

3.Show All users.

4.Show All groups.

5.Set user1 as group member of SQA.

6.Switch to user2.

7.Create a filename OSLabFile.txt and change its permission , read and execute only in group & user.

8. Delete user1

9. Delete SQA.

10.Change password of user1.